DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
SECOND SESSIONAL
SUBJECT: Machine Learning (CE622)

Examination	**: B.Tech Semester VI**	**Seat No.**	**:**
Date	**: 06/02/2023**	**Day**	**: Monday**
Time	**: 3:15 PM to 4:30 PM**	**Max. Marks**	**: 36**

INSTRUCTIONS:

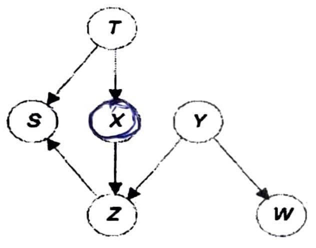
1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed. **[12]**

CO1 N (a) Which unsupervised learning technique would you use to categorize news articles into different categories? Justify your answer. [2]

CO2 R (b) Which are the limitations of K Means clustering algorithm? [2]

CO3 A (c) What is the Markov Blanket of node X for the Bayesian Network shown below? [2]



CO1 U (d) Define linearly separable patterns and non-linearly separable patterns with appropriate diagrams. [2]

CO2 U (e) i) 'Use predictions of multiple models as "features" to train a new model and use the new model to make predictions on test data .' Identify the ensemble learning model. [2]
ii) What are the functions used to combine outputs of several base estimators in Bagging Algorithm?

CO1 N (f) Explain the structure of a Multilayer Perceptron. Why do we need a differentiable activation function ? [2]

Q.2 Answer the following questions. [12]

CO2 N (a) i) What are the main components of a single layer perceptron? How are the outputs computed by a single layer perceptron? [2]
ii) Write and explain the Perceptron Learning Algorithm. State the condition under which the perceptron learning algorithm fails to converge. [4]

CO2 N (b) i) 'Ensembles tend to yield better results when there is diversity among the models.' Justify and state methods to achieve diversity in Ensemble learning models. [2]
ii) Explain the working of Random Forest Algorithm in detail. [4]

OR

Q.2 Answer the following questions. [12]

CO4 E (a) **i)** Consider a binary classification problem with input features 'psa level', 'age' and 'gleason score'. The class variable is 'cancer disease' which takes labels 1 or 0. The logistic Regression model generated upon training is: [2]
logit(cancer disease) = -6.3896 + 0.0266*(psa) -0.0208*(age) + 1.0790*(gleason score)
What is the predicted probability of having cancer disease=1 for a 69 year old man with a psa level of 10 and gleason score of 5 according to the given model. Show all the steps.

CO2 N **ii)** Write and explain the loss function used in Binary Logistic Regression. Can we use the Mean Square Error Loss function in Logistic Regression? Justify your answer. [4]

CO4 C (b) Consider the following dataset shown in Table Q.2 B. It consists of two features X1 and X2 and one output variable Y.An Adaboost classifier is applied on the given dataset. [6]
Let us suppose we chose the first model h1 with the following decision node (X1>2.0). This decision node **assigns class label (-1) to all samples which has value of X1>2.0 and class label (+1) to the rest of the samples.**
i) What is the initial weight assigned to all the samples?
ii) Based on predictions made by model h1, compute the error.
iii) Compute the performance of model h1.

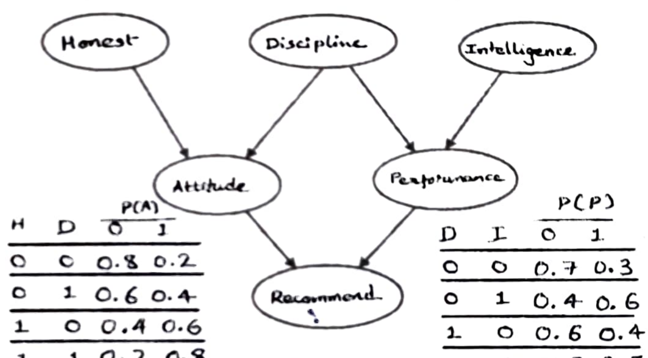
iv) Compute the updated weight of misclassified and correctly classified samples. Show which samples are misclassified.
Show all the steps and equations clearly.

X1	X2	Y
0.5	1.5	1
1.5	1.5	1
1.5	0.5	-1
2.5	1.5	-1
2.5	2.5	-1

Table Q.2.B

Q.3 Attempt ***Any TWO*** from the following questions. **[12]**

CO3 E (a) For the Bayesian Network shown in **Fig Q 2**, Using likelihood weighting technique, draw three samples and answer the following inference query, P(R=1 / H=0, P=1). Consider the following random numbers to draw samples, [25, 35, 45, 55, 65, 75, 85, 95, 35, 85, 45, 75] [6]



P(H)	0	1
	0.6	0.4

P(D)	0	1
	0.4	0.6

P(I)	0	1
	0.3	0.7

H	D	P(A) 0	P(A) 1
0	0	0.8	0.2
0	1	0.6	0.4
1	0	0.4	0.6
1	1	0.2	0.8

D	I	P(P) 0	P(P) 1
0	0	0.7	0.3
0	1	0.4	0.6
1	0	0.6	0.4
1	1	0.3	0.7

A	P	P(R) 0	P(R) 1
0	0	0.9	0.1
0	1	0.7	0.3
1	0	0.4	0.6
1	1	0.1	0.9

Fig: Q 2

CO3 E (b) For the Bayesian Network shown in **Fig Q 2**, Draw four samples using Gibb's sampling technique and answer the following inference query, P(R = 1/ I = 1, H = 0). Consider the following random numbers to draw the first sample, [25, 65, 75, 35] [6]

CO2 N (c) Describe the working of K-Means clustering algorithm for the data given below, [6]

Data	p1	p2	p3	p4	P5	P6	p7	P8
X1	2	4	6	3	5	7	8	1
X2	2	3	7	5	5	8	9	9

Initial cluster heads are C1 = p1 and C2 = p4.

DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
SECOND SESSIONAL
SUBJECT: (CE-618) NETWORK & INFORMATION SECURITY

Examination	**: B.Tech Semester VI**	**Seat No.**	: CE107
Date	**: 08-02-2023**	**Day**	**: Wednesday**
Time	**: 3:15 pm to 4:30 pm**	**Max. Marks**	**: 36**

INSTRUCTIONS:
1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed. **[12]**

CO1 U (a) Find any four points of elliptic curve E_{11} (1, 6). [2]
CO3 A (b) Find all primitive roots of 11. [3]
CO2 E (c) Justify: Rabin cryptosystem is not deterministic but RSA is deterministic. [1]
CO3 A (d) Apply Rail fence cipher to encrypt the plain text "computersoftware". [2]
CO2 E (e) Determine row and column number in decimal for S-Box1 to get the 4 bits output if the input to S-Box1 is 110110 in DES. [2]
CO2 N (f) Identify the types of elements present in DES round. Explain Feistel cipher structure. [2]

Q.2 Attempt *Any TWO* from the following questions. **[12]**

CO3 A (a) Find the value of 'x' for the following set of congruent equations [6]

$$X=3 \bmod 5$$
$$X=6 \bmod 7$$
$$X=4 \bmod 11$$

CO2 N (b) i. Give the mathematical proof of Elgamal cryptosystem regarding decryption and state why it is working in a correct manner? [2]
CO1 A ii. Given P(3,10), Q(9,7) for E_{23} (1, 1) Find R= P + Q. [4]
CO2 U (c) i. Explain Diffie-Hellman key exchange protocol. [2]
CO4 N ii. Explain Man in the middle attack in detail. [4]

Q.3 Attempt the following questions. **[12]**

CO2 C (a) Construct a diagram for general structure of Data Encryption Standard (DES). Explain each step in brief. [6]
CO3 A (b) Apply Hill Cipher to encrypt the plain text "abcd" using K and decrypt the cipher text using K^{-1} given below. [6]

$$K = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad K^{-1} = \begin{pmatrix} 18 & 17 \\ 17 & 18 \end{pmatrix}$$

OR

Q.3 Attempt the following questions. **[12]**

CO2 C (a) Construct diagrams for a round in DES (encryption site) and a DES function. [6]
CO3 A (b) Apply keyed transposition cipher to perform encryption of plain text "enemyattackstonightz" using encryption key={3,1,4,5,2}. Find decryption key and perform decryption of cipher text. [6]

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DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
B.TECH - Semester – VI (Computer Engineering)
SUBJECT: (CE 614)THEORY OF AUTOMATA & FORMAL LANGUAGE

Examination	: 2^{nd} **Sessional**	**Seat No.**	: CE107
Date	: **10/02/2023**	**Day**	:
Time	: **1. Hr. 15 Min.**	**Max. Marks**	: **36**

INSTRUCTIONS:
1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as Directed **[12]**

CO1-R **a.** Give recursive definition of δ^* for NFA **1**

CO2-A **b.** For any regular Language $L \subseteq \Sigma^*$, if the FA accepting L has at least n states. then every NFA accepting L has how many minimum number of states ? Justify your Answer. **2**

CO2-N **c.** In the proof for Kleene's Theorem - Part I (Regular Languages can be accepted by Finite Automata), consider this alternative to the construction described: Instead of a new state Qu and ^-transition from it to Q1 and Q2, make Q1 the initial state of the new NFA-^. and create a ^-transition from it to Q2. Either prove this works in general or give an example in which it fails. Credit will be given only if your answer and the justification/example, whichever applicable, is correct. **3**

CO1-N **d.** Identify the below given CFLs. **2**
- (1) S→ SaSbSaS | SaSaSbS | SbSaSaS | ^
- (2) S→A 1 A 1 A
  A → 0A | 1 A | 0 | 1

CO1-R **e.** Define: Pumping Lemma for Regular Languages **2**

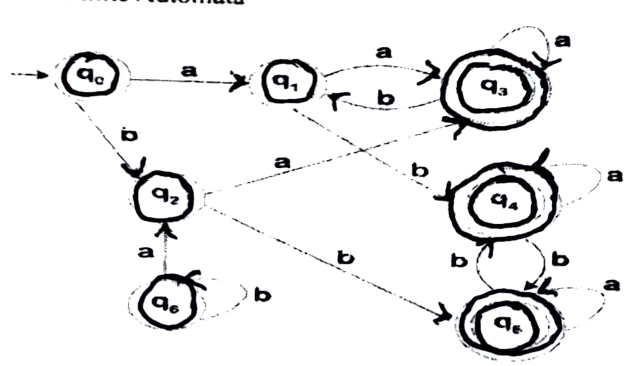
CO2-A **f.** Check whether the grammar given below is Ambiguous or not. **2**
S→ aaaaS | aaaaaaaS | ^

Q.2 Answer the Following (Any Two) **[12]**

CO1-U **a.** Using distinguishable strings, prove that the language of palindrome cannot be accepted by Finite Automata. **6**

CO1-N **b.** Write a CFG for $L = \{a^i b^j c^k \mid i \neq j+k\}$ **6**

CO1-A **c.** Minimize the Given Finite Automata **6**



Q.3 Answer the Following

CO2-A Prove the following Statement: Let $L \subseteq \Sigma^*$ is a language that is accepted by an NFA-^. Then, there is an NFA that also accepts L. **[12]**

OR

Q.3 Answer the Following

CO2-A Give a recursive definition of Regular Languages. Using the definition and structural induction, prove that any regular language can be accepted by a finite Automata. **[12]**

DHARMSINH DESAI UNIVERSITY, NADIAD
FACULTY OF TECHNOLOGY
SECOND ~~FIRST~~ SESSIONAL
SUBJECT: COMPUTER NETWORKS

Examination	**: B.Tech Semester VI**	**Seat No.**	: 107
Date	**: 07/02/2023**	**Day**	**:Tuesday**
Time	**: 3:15pm to 4.30pm**	**Max. Marks**	**: 36**

INSTRUCTIONS:
1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

Q.1 Do as directed. **[12]**

CO3 A (a) In an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following can be used for this? [2]
(1) X sends an ARP request packet with broadcast IP address in its local subnet
(2) X sends an ARP request packet to the local gateway's MAC address which then finds the MAC address of Y and sends to X
(3) X sends an ARP request packet with broadcast MAC address in its local subnet
(4) X sends an ARP request packet to the local gateway's IP address which then finds the MAC address of Y and sends to X

CO4 R (b) Which protocol(s) is/are used by DNS and FTP at transport layer and network layer? [2]

CO3 A (c) ICMP can not be used to determine the round-trip time of packets between network devices. Is the given statement true/false? Justify your answer. [2]

CO5 U (d) Discuss "milk" and "wine" policies of the load shedding. [2]

CO2 A (e) For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. Calculate the minimum time required to transmit the data in seconds. [2]

CO1 N (f) Below are the link state packets generated by routers in a subnet. Construct the subnet graph. [2]

A		B		C		D		E		F	
Seq		Seq		Seq		Seq		Seq		Seq	
Age		Age		Age		Age		Age		Age	
B	4	A	4	B	2	C	3	A	5	B	6
E	5	C	2	D	3	F	7	C	1	D	7
		F	6	E	1						

Q.2 Attempt ***Any TWO*** from the following questions. **[12]**

CO3 A (a) How many maximum data bytes can be accommodated in a UCP datagram? [6]
How many maximum data bytes can be accommodated in a TCP segment?
Why is initial sequence number random in TCP?
Suppose TCP has a file of 10000 bytes to transfer. The sequence number of the first segment is 1104. If the size of a segment is same as the maximum transfer unit in Ethernet, what will be the sequence number of the next two segments? Which acknowledgement will be generated by receiver if the first segment reaches after the second one?

CO1 E (b) (i) Correct the following module of go back n protocol. [6]

```
case timeout:
next frame to send = frame_expected;
for (i = 1; i < nbuffered; i++)
 {
s.info = buffer[];
s.seq = next_frame_to_send;
s.ack = (ack_expected + MAX SEQ) % (MAX SEQ+1);
to physical layer(&s);
start timer(s.seq);
inc(next_frame_to_send);
}
```

(ii) Explain following function with examples (for selective repeat protocol) if a 3-bit sequence number is used.

```
static Boolean between(seq_nr a, seq_nr b, seq_nr c)
{ if (((a<=b)&&(b<c)) || ((b<c)&&(c<a)) || ((c<a)&&(a<=b)))
        return(true);
  else
        return(false); }
```

CO3 U (c) Which of the fields of TCP header can have zero value? Explain flow control of TCP. [6]

Q.3 Answer the following questions. **[12]**

CO3 A (a) Briefly discuss the leaky bucket algorithm. The following packets arrive at a switch that uses a leaky bucket algorithm. The bucket can contain up to 3000 bytes. **[6]**

Packet No.	Arrival time slot	Size(Bytes)
1	0	100
2	1	400
3	2	400
4	3	1000
5	4	1000
6	5	1000
7	6	600
8	7	1100
9	8	1000

The leaky bucket operates on packets in FIFO manner, and can send 1 packet in every 3 time slots (i.e. bucket outputs a packet at slot 0, slot3, slot 6 so on). Assuming no packet arrives after 8^{th} slot, show when packets leave and which packets are dropped (if any).

CO4 C (b) Consider an IP packet with a total length of 250 bytes and an id field equal to 17, sent from a host A to host B, passing through routers X and Y respectively. Assume that the subnet where host A is connected has an MTU of 500 bytes, the subnet where host B is connected has an MTU of 80 bytes and the subnet between X and Y has an MTU of 120 bytes. Assuming that the "Don't fragment" flag is not set, how many fragments does router X divide the packet into? Show the total length, Identification, MF and fragment offset fields of the IP header in each packet transmitted over the link from X to Y. **[6]**

OR

Q.3 Answer the following questions. **[12]**

CO3 A (a) (i) Consider the subnet of **figure-1**. Distance vector routing is used, and the following vectors have just come in to router C: from B: (5, 0, 8, 12, 6, 2); from D :(16, 12, 6, 0, 9, 10); and from E: (7, 6, 3, 9, 0, 4). The measured delays to B, D and E are 6, 3 and 5, respectively. Consider the destination order in routing table (A, B, C, D, E, F). What is C's new routing table? Give both, the outgoing lines to use and the expected delay. **[4]**

(ii) Describe count-to-infinity problem of distance vector routing. **[2]**

CO4 C (b) Consider the subnet shown in **figure-2**. A sink tree for router *I* is shown in **Figure-3**. **[6]**

(i) How many packets are generated by a broadcast from router *I* using Spanning tree based broadcast technique?

(ii) Build a tree generated by reverse path forwarding for a broadcast from router *I*.

(iii)How many packets are generated by a broadcast from router *I* using Reverse path forwarding broadcast technique?

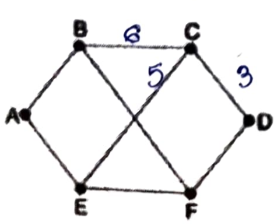


Figure-1

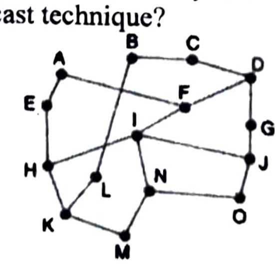


Figure-2

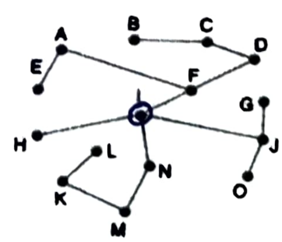


Figure-3